BEFORE DAY 1

Pre-Flight Setup

TouchDesigner · 5-Day Course

Finish all six steps at least 72 hours before our first session.

We have five hours together. None of them can be spent on installers.

Step 0 — Will your machine cope?

Check this before you download anything.

Windows

- Windows 10 or 11, 64-bit
- A dedicated NVIDIA GPU is strongly preferred
- Intel/AMD integrated graphics will run Days 1–2, but may struggle on Day 3
- 8 GB RAM minimum, 16 GB comfortable

macOS

- Apple Silicon (M1 or newer) is what Derivative recommends
- Intel Macs need a discrete AMD GPU
- Older Intel Macs with integrated graphics: tell me now, I will pair you up

You also need

- A working webcam — built-in laptop cameras are completely fine
- About 3 GB of free disk space (the MediaPipe plugin alone is over 500 MB)
- A room where you can wave your hands around in reasonable light

1 Install TouchDesigner

Go to the official download page:

derivative.ca/download

Choose the installer for your operating system (Windows or macOS).

Install the build we are standardising on:

2025.33070

Run the installer and accept the defaults. Nothing here needs customising.

Open TouchDesigner once to confirm it launches. You will see a licence window — that is Step 2.

Same build, everyone

The files I hand out are made in 2025.33070. A different build may refuse to open them, and on a five-hour course that is unrecoverable. Please do not install "the newest" instead.

Official install guide

If anything is unclear, Derivative's own walkthrough is good:
the Getting Started guide

2 Get your free licence key

We use the free **Non-Commercial** licence. It costs nothing and does not expire.

Create a free account at **derivative.ca** — every new account comes with Non-Commercial keys.

Open TouchDesigner. In the licence window, sign in with that account.

Select your Non-Commercial licence and click **Create Key**.

You are done when TouchDesigner opens with no licence pop-up.

The one limit

Non-Commercial caps image resolution at 1280×1280. Everything we build fits inside that, so it will not affect the course — but it is why our project runs at 1280×720 rather than full HD.

Students & faculty

An Educational licence exists too, but you do not need it for this course. Details: docs.derivative.ca/Licensing

3 Download the MediaPipe plugin

This is the piece that lets TouchDesigner see your hands. There is nothing to install — it is a folder you unzip.

Go to the releases page:

github.com/torinmb/mediapipe-touchdesigner/releases

Find the newest release — currently **v0.5.3**.

Under Assets, download **release.zip**.

Unzip it somewhere permanent — a course folder on your Desktop, not Downloads.

It is a big download. Do this on decent wifi, not ten minutes before class.

Do NOT download Source code

GitHub offers "Source code (zip)" right underneath. That is the developer version and it will not work. You want release.zip and nothing else. This is the mistake almost everyone makes.

What it is

A free, open-source plugin by Torin Blankensmith. Runs on Mac and PC. Read more: the project README

4 Run the MediaPipe example

Inside the unzipped folder, open **MediaPipe TouchDesigner.toe**.

Wait. The first launch takes a while — it starts a small local web server and loads the tracking models. A loading screen for thirty seconds is normal.

Allow camera access if your browser or system asks. Then find the hand tracking example and hold your hand up to the camera.

Success looks like

Dots and lines drawn over your hand, following it as you move. Numbers changing on screen when you open and close your fingers.

Keep the folder together

Do not move MediaPipe.tox out of the folder or rename anything. It is over 500 MB and the project expects to find it exactly where it is.

5 Sort out camera permissions

A blocked camera looks identical to a broken install. Check this even if everything already works.

macOS

System Settings → Privacy & Security → Camera → switch TouchDesigner ON. You may need to quit and reopen TouchDesigner afterwards.

Windows

Settings → Privacy & security → Camera → turn on camera access, and make sure desktop apps are allowed to use it.

Only one app can use the camera at a time

Zoom, Teams, Meet, Photo Booth, the Windows Camera app, or a browser tab on a video call will all lock TouchDesigner out — and TouchDesigner will just show black, with no explanation.

This matters during class: if you join our session on a laptop and TouchDesigner needs the same camera, one of them loses. Join the call from your phone or a second device, or be ready to drop video while you build.

6 Post your proof-of-life screenshot

Your webcam feed

A TouchDesigner window showing live video of you.

Live hand tracking

The tracking dots drawn on your hand, or the tracking numbers visibly changing.

Where and when

- Post both screenshots to our shared channel, with your name, operating system, and TouchDesigner build number.
- **Deadline: 48 hours before Day 1.**
- If something is broken, post the screenshot of what you DO see plus the error text. A stuck student who posts early gets fixed. A stuck student who waits loses Day 1.

If something goes wrong

Work down this list before asking — it solves most cases.

Camera shows black

Another app has it. Quit Zoom, Teams, browser tabs, the Camera app — then restart TouchDesigner.

Camera dropdown is empty

Permissions. Go back to Step 5. On Windows, check that desktop apps are allowed.

MediaPipe never finishes loading

Give it a full minute. Then close it, make sure no other app has the camera, and reopen. If it persists, you may have downloaded the source code instead of release.zip.

Everything is tiny and blurry

A resolution mismatch, and it is a real TouchDesigner quirk rather than your mistake. Bring it to class — it is a two-minute fix and a good lesson.

TouchDesigner will not open my file

Almost always a build mismatch. Check Help → About for your build number and compare with 2025.33070.

My laptop has no webcam

Tell me now. I will send you a recorded tracking file so you can follow along identically.

Optional: warm up before Day 1

Not required. Nothing here will be assumed on Day 1 — we start from zero.

Derivative's official docs

The main hub for everything TouchDesigner

docs.derivative.ca

Getting Started guide

Derivative's own beginner walkthrough. A gentle first look at the interface.

docs.derivative.ca/Getting_Started_With_TouchDesigner

The MediaPipe plugin README

Explains what the plugin can track — hands, faces, bodies, objects. Worth a skim for ideas.

github.com/torinmb/mediapipe-touchdesigner

Your checklist

Tick all six, then post your screenshots.

- [ ] TouchDesigner 2025.33070 installed and opening without a licence pop-up
- [ ] Free Non-Commercial licence key created and active
- [ ] release.zip downloaded from the MediaPipe releases page and unzipped somewhere permanent
- [ ] MediaPipe TouchDesigner.toe opens and tracks my hand
- [ ] Camera permissions granted, and no other app is holding the camera
- [ ] Both screenshots posted to the shared channel with my name, OS and build number